

The Efficacy of Voluntary Overcompliance for Decarbonization: Evidence from California

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Abstract

Overlapping environmental policies can be net beneficial if they yield additional, cost-effective mitigation. I study this overlap in the Californian context, where aggressive decarbonization regulation exists alongside high participation in voluntary green power providers called Community Choice Aggregators (CCAs). I find that CCAs procure more green power for their customers and exhibit measures of higher willingness-to-pay for decarbonized power. However, CCAs amount to a reshuffling of voluntary greenness rather than additionality at a statewide level, suggesting that while voluntary green power programs can advance participants' decarbonization goals, overall system benefits can be undermined by dynamic responses to voluntary green participation.

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I. Introduction

Substituting low-carbon generation resources for fossil resources is key to lowering the emissions intensity of electricity supply. Many mechanisms can facilitate this transition, such as renewable portfolio standards (RPS), regulatory mandates that impose minimum levels of renewable energy content for electricity providers (Barbose et al. 2016); incentives or subsidies to stimulate investment (Aldy et al. 2023); competitive procurement auctions (Haufe and Ehrhart 2018); cap-and-trade emissions schemes (Schmalensee and Stavins 2017); carbon taxes (Timilsina 2022); and individuals engaging in “self-regulation” via voluntary purchases of renewable power (Lyon and Maxwell 2008). Although a carbon tax is widely considered the most economically efficient policy, political economy challenges make practical implementation difficult. Thus, programs such as RPS are more commonly implemented despite being relatively expensive forms of abatement, and policymakers sometimes employ multiple approaches simultaneously. At times, overlapping policies enable additional, cost-effective mitigation; in other cases, they may instead prove “redundant but harmless, or working at cross- purposes” (Levinson 2012). This paper examines whether the interactions between top-down regulatory standards and bottom-up voluntary overcompliance are beneficial for emissions reduction efforts. Voluntary overcompliance, or voluntary green power programs, enable renewable energy purchases on behalf of consumers that exceed policy-mandated minimum percentages (Sumner et al. 2023). Sufficiently large voluntary green power demand would signal the wholesale sector to increase renewable energy supply and thereby boost decarbonization progress.

Twenty-nine US states and the District of Columbia have an RPS, and in all US states, some customers are offered forms of voluntary green power (although in a handful of states fewer than 100 customers purchase this service) (NREL 2023). A variety of institutional arrangements, financial products, and government programs can facilitate voluntary green power.¹ One such model is Community Choice Aggregation (CCA), where local government enters into the electricity retailing space and procures power for its member communities. CCAs may engage in voluntary greenness by procuring RPS-eligible² resources in excess of the

¹ The National Renewable Energy Laboratory defines six categories of voluntary green power procurement: utility green pricing, utility renewable contracts, competitive suppliers, unbundled renewable energy credits (RECs), Community Choice Aggregators (CCAs), and Power Purchase Agreements (PPAs).

² The RPS-eligible renewable energy sources are bioenergy, geothermal, small hydroelectric, solar, and wind. California recognizes nuclear and large hydroelectric plants as providing carbon-free power but at this time does not count them as RPS-eligible.

mandated floor. CCAs exist in ten US states with four more considering authorization, and a majority of CCAs offer green options that exceed the RPS. In this paper, I examine the case of California, which has both an aggressive RPS and a high degree of participation in CCAs.

I empirically test the renewable procurement performance of CCAs relative to the RPS minimum. My model supports the hypothesis that CCAs engage in voluntary overcompliance by procuring more green power than required by the RPS, with CCA formation and participation in voluntary overcompliance strongly correlated with indicators of high willingness-to-pay for renewable energy. However, I find that the net effect is not meaningfully different from a binding RPS alone, nor is it clearly additional to pre-CCA levels of voluntary overcompliance. The lack of additionality arises for three reasons. First, as the RPS has risen over time so too has the marginal cost of compliance, which increases the difficulty of consistently exceeding mandated percentage levels. Second, when load departed to CCAs, incumbent utilities responded to the resulting over-procurement of renewables and lost retail sales by shedding older, relatively expensive renewable contracts; these declines offset the gains from overcompliance by high-performing CCAs. Third, CCAs had no positive impacts on other publicly-owned or competitive retail providers in the sector, because they do not meaningfully compete with those entities.

This paper contributes to the literature on overlapping quantity policies for climate regulation (Levinson 2012, Goulder and Parry 2008, Fischer et al. 2017). Interactions between policies have often proved problematic as they raise total costs with little accompanying additional reduction in emissions. While prior work has focused on the interplay between national and state policy (Goulder and Stavins 2011), this research extends the framework to the dynamics between state and local government. As policymakers search for instruments to rapidly achieve emissions reductions, voluntary overcompliance may indicate that additional cost-effective mitigation is possible. Because is the cumulative rather than marginal output of emissions that primarily determines damages from climate change, incremental progress, especially in early years, results in greater levels of avoided harm overall.

This paper also contributes to the literature on voluntary versus mandated action to ameliorate a market failure. A belief that the regulator will soon raise the standard may lead firms to hedge against the change by building a buffer (Lyon and Maxwell 2004); anticipation of future shocks to performance can have a similar effect (Toffel and Short 2011). Voluntary overcompliance may indicate a non-binding standard, implying that additional progress can be

achieved at low cost. Firms may also exceed standards to signal pro-environmental values for reputational gains (Maxwell et al. 2000, Arora and Gangopadhyay 1995). Voluntary overcompliance has been documented in the contexts of environmental regulation, such as the US EPA's 33/50 program on reducing toxic chemical emissions (Arora and Gangopadhyay 1995, Khanna 2001), and corporate strategy, such as environmental, social and governance (ESG) investing and carbon disclosure (Lyon and Maxwell 2008, Hsueh 2019, Duchin et al. 2022). Findings are mixed as to whether such self-regulation measures actually translate to meaningful declines in emissions. I look at the performance of voluntary overcompliance in the setting of local governance and electricity sector decarbonization. Relative to the private sector, pro-environmental signaling may take on more potency for public power providers because its decision-making processes are influenced by the preferences of its constituents. Political support for decarbonization can lead to positive demand for green power beyond the RPS minimum; this political preference influences elected officials who administer the electricity retailer to procure more green power than regulation requires, and also implies that the community's consumers have a higher willingness-to-pay for green power. This aligns with prior empirical evidence that, just as people "vote with their feet" with respect to taxes or bundles of public goods, they also do so with respect to environmental quality (Banzhaf and Walsh 2008).

The work also ties in to a literature studying the effects of environmental regulation on decision-making for participants in electricity markets. The impact of pollution permits on firms' generation and investment strategies is a longstanding topic of study (Laffont and Tirole 1994, Fowlie 2010), as is measuring the cost of carbon pricing schemes on energy prices (Borenstein and Kellogg 2022, Holland et al. 2022). The majority of this work, however, focuses on effects in the wholesale market. This study instead focuses on the implications of clean energy mandates for procurement decisions by retail providers. Retailer actions are important because demand preferences can influence supply decisions, i.e., the increase in demand for green power can induce an upstream expansion in the supply of green power.

California's particular CCA design has two features that present a uniquely strong opportunity to observe the impact of a voluntary green power mechanism on overall decarbonization performance. First, California's CCAs often tout environmentally-progressive reputations: all except one offer voluntary green power options, and many have configured their default offering to overcomply with the RPS. Second, California's CCAs are opt-out rather than

opt-in, and the vast majority of customers do not take action to opt out. This is consistent with prior literature that shows a large default effect in electricity retail pricing programs (Hortacsu et al. 2017, Fowlie et al. 2021), and the broader literature on default design (Thaler and Sunstein 2008). CCAs with high-renewable default portfolios therefore give rise to a large demand for green products, as they capture both customers with high willingness-to-pay for voluntary green power and customers who are inattentive to the clean energy content of their power.

Prior research on CCAs has explored sociopolitical aspects of their formation, such as coalition-building (Hess 2019), and diverse community motivations for establishing a CCA (Gunther and Bernell 2019). Other studies have indicated that CCAs procure more clean electricity than is required by California's RPS (Trumbull et al. 2019, 2020), but the net effect of CCA overcompliance on statewide adoption of renewable energy had remained unclear. Determining whether CCAs' voluntary overcompliance indeed accelerated decarbonization efforts requires a careful definition of additionality. In the context of this study, regulatory compliance means a provider has met the RPS in a given time period. Voluntary overcompliance means the provider has procured more than the mandated quantity in a given time period. That is, *any* level of voluntary greenness, regardless of which provider procured it, represents additional progress relative to an RPS target. Thus, if the question is quantifying CCAs' total voluntary overcompliance, this is simply the difference between CCAs' proportion of renewable energy and the RPS floor. But if the question is instead whether CCA contributions are *additional*, this requires comparison to whatever existing levels of voluntary greenness were present due to the actions of other providers. Though CCAs have often been lauded as agents of "energy democracy" and "energy equity" to promote green power access for communities (Hess 2019; Taminiau and Byrne 2025), I observe that incumbent utilities also engaged in voluntary green procurement before large-scale CCA entry. In fact, CCA expansion is correlated with a reduction in California's overall levels of voluntary green procurement. This short-term erosion of progress is unsurprising in the face of an increasingly stringent RPS and macroeconomic shocks such as the Covid-19 pandemic. But I show that it is also consistent with an economically-rational response to successful CCA expansion. My findings demonstrate that CCAs can be effective as a matching mechanism, by placing the greatest levels of renewable energy procurement in the hands of those most politically, economically, and/or socially willing to take it on. However, the

CCA model's success can also be associated with a chilling effect, at least in the short term, on the rest of the sector.

The paper proceeds as follows. Section II introduces the empirical setting. Section III models the community characteristics associated with voluntary green procurement, and Section IV analyzes additionality in terms of net renewables procured. Sections V and VI assess CCA impacts on retail switching and improved investment in renewable generation respectively. Section VII discusses policy implications and Section VIII concludes.

II. Background and Context

A. Portfolio Choice and Retail Choice

Voluntary green portfolio offerings can be, but aren't necessarily, features of competitive retail electricity markets. In the US, the structure of the retail electricity market differs by state, ranging from zero retail choice to full retail choice. Even in fully liberalized settings, competitive retailers might offer consumers the choice of different rate structures but no option to select the clean energy content of the power being procured on their behalf. In this case, customer preferences concerning clean energy will not carry over into sending a signal to the wholesale generation sector. Conversely, a non-competitive retail sector might still contain a monopolist that offers customers a choice between a default procurement portfolio and paying a premium to receive a portfolio with a larger proportion of clean energy. Therefore, this work focuses on the effects of portfolio heterogeneity—rather than price heterogeneity—on the wholesale sector. This could have implications for the structure of the retail sector but it is not the same as discussing retail competition in the classic sense.

Notwithstanding the ambiguous effect of market structure on the supply of green power, the literature on retail competition still provides a useful theoretical framework. The effectiveness and value of competitive electricity retail markets have been debated since the concept was introduced. Joskow (2000) expresses skepticism, as competitive retailers are still inherently tethered to upstream regulated network monopolies, which constrain opportunities for meaningful innovation or cost saving. Though retailers could theoretically provide other value-added services, Joskow concludes that such opportunities are likely small. Littlechild (2000), in contrast, argues that greater competition on the demand side could induce improvements at the wholesale level, and result in better cost-minimization than systems without retail competition.

Twenty years after his original argument in favor of retail competition, Littlechild (2021) asserts that sophisticated demand-side engagement is crucial for successful deep decarbonization, and that this emphasizes the importance of expanding and innovating in retail competition more than ever. Empirical literature assessing the performance of retail competition has tended to focus on measuring price impacts, in most cases finding that the introduction of retail competition into liberalized electricity markets has yielded few pricing benefits to consumers (e.g., Joskow 2006, Su 2015). This study measures how the average renewable energy content has changed with the introduction of heterogeneous procurement portfolios into a market. A complete assessment of the modern electricity retailer should account for performance in terms of clean power and not just cost-competitive power. If certain retailer models are shown to help accelerate progress to a decarbonized wholesale electricity supply, this is a substantial value-added service.

B. California's Various Load-Serving Entities

California's electricity retailing sector has a complex history. The state introduced a competitive retail market in the late 1990s, but abruptly reversed course in the midst of the 2001 energy crisis, when customers were largely returned to the default service provider for their geographic region. A given household generally had no choice over their retailer.³ Because a CCA arises within the territory of an incumbent private electricity provider, the entry of CCAs over the past fifteen years substantially expands the total number of electricity retailers in the state and re-introduces some choice of retail provider for households within their jurisdiction.

As of 2021, 92 separate Load Serving Entities (LSEs) across five provider types operated as retail electricity providers in California. These LSEs vary greatly in terms of service territory size, types of customers served, governance structure, and regulatory oversight. Summary statistics for the five categories of LSE—IOWs, CCAs, publicly-owned utilities (POUs), direct access providers (DAPs), and co-ops—are given in Table 1. This paper focuses on the dynamics between two of these types, IOWs and CCAs.

IOWs are private, regulated utilities. Historically IOWs were vertically integrated into all segments of the industry: generation, transmission, distribution and retail sales. Though

³ Some commercial and industrial customers can take service from alternative retailers through California's Direct Access Providers (DAPs). However, the total allowable DAP load is capped, so in practice these customers also experience barriers to switching away from incumbent providers (CPUC Staff 2021).

deregulation in the 1990s forced California's IOUs to largely divest from generation and to allow other entities to enter transmission and retail, they still do own some power producing assets as well as a significant portion of the transmission infrastructure across the state, and they can continue to perform distribution and retail functions within their service territories. California has three major IOUs (PG&E, SCE, and SDG&E) and three Small/Multijurisdiction IOUs (Bear Valley Electric Services, Liberty Utilities (CalPeco Electric), and PacifiCorp (Pacific Power)). These latter three amount to ~1.5% of IOU load. Because they are very small and subject to special regulations, and because the latter two also serve a large number of customers outside of California, this study focus only on the three large IOUs.

AB117, passed in 2002, enabled local governments to become new participants in energy procurement by forming a community choice aggregator (CCA). CCAs are public entities formed via Joint Powers Agreements (JPAs) between one or more cities or counties. CCAs may only form within the existing service territory of an IOU, and while a CCA purchases power on behalf of the customers living in its geographic region, it still uses the transmission and distribution network of the incumbent IOU. Thus, any given household in IOU service territory can select from at most one CCA and one IOU as its electricity supplier and has no choice over its provider of distribution service. Unlike traditional competitive retailers that must entice customers to switch from a default provider, CCAs in California are opt-out entities, so they avoid the marketing costs generally associated with successful startup of new retailers. California's CCAs have experienced rapid growth and high market penetration. The first operational CCA launched in 2010, and by 2021 over eleven million customers participated in a CCA (>20% of load). As some CCAs procure very high levels of renewable power, the emergence of these "leadership communities" may indicate a decarbonization benefit to allowing the formation of new retailers.

But the assertion that CCAs are "accelerating the state's transition" (Trumbull et al. 2020) does not account for key facts about the performance of incumbent utilities. First, it assumes that the right baseline for comparison is the RPS minimum: if actual progress in the absence of CCA entry was under or over the RPS level, reported CCA impacts may be distorted either in the positive or negative direction. Second, it does not sufficiently account for incumbents' rational responses to CCA growth. CCAs' voluntary procurement meaningfully accelerates progress if and only if it is both larger than incumbents' levels of voluntary

procurement as well as large enough to offset any negative changes in incumbents' procurement strategies in response to CCA entry. As I show in section IV.B., CCA voluntary overcompliance has been accompanied by other short-term setbacks—declines in both incumbents' renewable procurement and overall state voluntary overcompliance.

C. Renewable and Clean Energy Regulation in California

California has multiple, overlapping decarbonization policies. The current law driving economy-wide decarbonization is SB 32, which directs the state to reduce GHG emissions by 40% below 1990 levels by 2030. The program that facilitates these reductions is cap-and-trade (CAT), with its associated system of tradable emission allowances. California's SB 100 is a separate law focusing on decarbonizing the energy sector. It "established a landmark policy requiring renewable energy and zero-carbon resources supply 100 percent of electric retail sales to end-use customers by 2045" (CEC 2024b). The primary mechanism for achieving SB 100 is the Renewable Portfolio Standard (RPS), which "sets continuously escalating renewable energy procurement requirements for the state's load-serving entities" (CEC 2025), and which uses a system of tradable renewable energy certificates (RECs) to track compliance. Note that voluntary allowance retirements under CAT are not the same as RPS voluntary overcompliance.

The state anticipates that electricity decarbonization will be achieved by ratcheting up the minimum required percentage of retail purchases accounted for by clean energy, altering retailers' power procurement decisions and thereby indirectly stimulating greater investment in renewable generation. LSEs were required to achieve 33% renewable energy by 2020 and will need to reach 60% by 2030. In the California regulatory context, *renewable* energy refers to biofuel, geothermal, eligible (small) hydro, solar, and wind generation. Nuclear and large hydroelectric generation are zero-carbon resources but are not considered renewable. The sum of renewable, nuclear, and large hydro is the total amount of *clean energy* procurement in each portfolio. I report LSE performance with respect to renewable energy in the main text and total clean energy in the Appendix. The remaining specified energy fuels are classed as fossil carbon resources: coal, natural gas, and a residual "other" category that encompasses such fuels as petroleum, diesel, and propane. Finally, unspecified power "refers to electricity that is not traceable to specific generation sources by any auditable contract trail or equivalent" (Public Utilities Code Section 398.2). Unspecified power indicates the extent to which an LSE buys off

the short-term market rather than contracting for power. While the exact composition of unspecified power may not be known, state calculations find that the average emissions intensity of unspecified power is slightly higher than that of a typical natural gas plant in California.

The details of determining RPS compliance are important to understanding the dynamic behavior I observe. The RPS is a percentage-based standard that is assessed on a per-LSE basis. Compliance is assessed for multi-year compliance periods (CPs). A retailer can be out-of-compliance for some of the years within the CP as long as the overall period average is sufficient. A retailer facing a large decline in load can decrease its renewable procurement in absolute terms but maintain compliance on a percentage basis. A retailer is allowed to offer a 100% renewable product and an out-of-compliance product as long as its average proportion of renewable energy exceeded the RPS floor.

Environmental policy at the local government level can impose more specific or stringent conditions than state policy. Cities can implement decarbonization measures through renegotiations of their franchise agreements with their local utility (Cook et al. 2020). For example, the city of San Diego recently renegotiated its franchise agreement with its private utility provider (SDG&E), whereby it agreed to establish a \$20 million climate equity fund and \$10 million clean energy access fund (Nikolewski 2021). Cities can also implement local clean power ordinances, such as San Francisco's requirement that large commercial buildings must use 100% clean power (San Francisco Ordinance No. 220-19). Just as the RPS is one part of a broader state policy environment, CCA formation is but one tool that a local government can use to further its decarbonization goals.

III. CCAs and Preference Matching

It is trivial to verify that the CCA-wide average proportion of renewable energy is higher than the IOU-wide average (Figure 1). The empirical exercises discussed in this section focus on understanding who tends to form CCAs and whether any community traits predict higher performance. A given LSE will partially derive its preferences for green power from preferences of its constituents, but will also be influenced by the institutional environment in which it operates. This includes regulatory constraints as well as the nature of political oversight for that LSE. I estimate the associations between a variety of attributes that might indicate higher willingness-to-pay for decarbonized power, and the proportion of decarbonized power present in

various LSEs' default portfolios. I find that communities forming CCAs and receiving more decarbonized power tend to be higher income and politically supportive of climate change mitigation policy.

A. Methodology

I estimate reduced-form equations to determine the effects of LSE type and community characteristics on the equilibrium quantity of green electricity procured on behalf of retail customers. This is accomplished via two modal search exercises. The first is a binomial logit model of CCA formation to gain insight into the community characteristics associated with a desire to access a mechanism for voluntary green procurement. The second regresses green power on community characteristics to determine whether participation in a CCA and indicators of higher willingness-to-pay correlate with the actual intensity of decarbonized power procured.

The logistic regression takes the form:

$$is_CCA_i = X'_i\beta + \epsilon_i$$

where each i represents a community, X is a vector of community characteristics, and ϵ is an idiosyncratic error term. The outcome variable is_CCA_i takes on the value one if the community became served by a CCA in any year between 2010 and 2020. A range of covariates that might indicate higher propensity for CCA formation, including measures of income, political preferences, relative cost-of-service, and presence of local renewable resources that could provide local economic benefits, are considered so as to attenuate omitted variable bias. A “community” is defined as the unit of decision-making over a default electricity provider, which generally corresponds to a municipal jurisdiction (i.e., an incorporated city, an incorporated town, or the collective census-designated places that comprise the unincorporated areas of a given county). It is an administrative definition rather than a natural one. For example, Calabasas is an incorporated city in Los Angeles County, and so it counts as a community in the dataset. Altadena, which has a town council and a larger population than Calabasas, is not incorporated and so it is aggregated along with all the other census-designated places in the county into a single “Unincorporated Los Angeles County” observation. The model uses all communities eligible for CCA formation, that is, all communities originally within the network service territory of an IOU. I exclude any community served by POUs because these communities

cannot form a CCA that enters into competition with their local POU. Communities that were in the process of forming a CCA at the end of the observed period are recorded as served only by an IOU. This group includes communities in which a CCA became operational after 2020 and other communities that abandoned the process of CCA formation in that period. A community where a CCA was formed but then went bankrupt is coded as having a CCA.

Access to or participation in an alternative retailer does not automatically mean the community must be “greener” than others. The purpose of the second regression model is to verify whether the community characteristics that correlate with CCA formation also significantly correlate with the intensity of decarbonized energy actually procured for that community. In the OLS regressions, the outcome variable of interest is the percentage of a certain type of energy (renewable energy or overall clean energy) in the default electricity portfolio for a given community. The renewable energy regression takes the form:

$$pct_renewable_i = X'\beta + \epsilon_i$$

where each i is a community, and the vector X contains variables measuring community characteristics and size of the LSE. The clean energy regression (see Appendix) is of the same structure.

The goal of this regression is to assess which community characteristics are important determinants of decarbonized energy procurement, and whether those characteristics have differing effects depending on the type of LSE the community belongs to. So, this model encompasses all communities served by an IOU, POU, or CCA, which have service territories that can be mapped to geographic areas.⁴ The community characteristics considered here are the same as those in the logit regression, with two additions. First, I include a covariate for the size of the LSE, to test the effect of economies of scale on decarbonized energy procurement. Second, I include a public/private governance fixed effect. Consider two identical sets of communities, where one set is served by a public power provider (i.e., CCA or POU) and the other is served by a private utility (IOU). Their procurement portfolios may look substantially different due to divergent political and economic incentives. The IOU’s chosen portfolio may account for

⁴ I exclude the DAPs for two reasons. First, DAP customer identities are confidential but are usually large commercial/industrial loads, where pure cost considerations are more likely to govern procurement decisions. Second, a DAP is not restricted to serve a specific community (like a CCA or POU) or region (like an IOU). Therefore, it is not feasible to include them in a regression of renewable energy use on community characteristics.

customer preferences somewhat, in terms of relevant economic interest, but ultimate accountability is to shareholders. In contrast, in public power settings, induced preference theory and positive responsiveness predict that constituent preferences are relatively more salient determinants of decisions such as power procurement. Indeed, Wald tests show that the CCAs and POUs pool in the renewable energy regression but are distinct from the IOU group (see Appendix for detail).

The goal of these regressions is to understand CCAs as the product of a matching mechanism between communities that already have high willingness to pay for decarbonized power and increased intensity of decarbonized power received—not whether community characteristics causally altered procurement patterns. Therefore, the analysis is not threatened by identification problems involving regression of endogenous variables. Due to clustering at the community level, I cannot detect variation within a given community. As this means a coarsening of information about heterogeneity in consumer preferences, any matching I do find is likely an underestimate.

B. Data

In most of California, a single LSE is the default provider for a set of cities, towns, or counties. The decisions to create a CCA and to select a default electricity portfolio are made by a local government, not an individual customer. Therefore, data were collected for each city and town in California. In the case of unincorporated communities, the choice of electricity provider is made by a county government such that the same LSE will serve all unincorporated portions of that county. So, data for all census designated places within a given county were aggregated into a single observation per county. This yields 482 cities or towns + 57 counties containing unincorporated communities for a total of 539 observations. The number is reduced to 523 after dropping observations with missing data. Of these, 470 are IOU or CCA communities and used for the binomial logit estimation. Although communities are of heterogeneous sizes, this does not result in a clustering problem because the outcomes in question and level of decision-making are both at the community level (i.e., whether the community joined a CCA, and what the community decided should be the percentage of decarbonized energy they receive by default), so it is appropriate to use a community as the unit of analysis.

Data were compiled for eighteen parameters describing socioeconomic, demographic, and other characteristics of each community that plausibly could indicate higher willingness to pay for green power. The main variables of interest are income, various measures of political preference, and LSE size. Table 2 defines the full set of variables used in developing the regressions. Data on electricity procurement by type of generation facility are from the Power Source Disclosure (PSD) filings by each LSE in 2020, from which are calculated weighted averages of the percent of electricity from sources that are renewable and the percent from clean energy sources. For example, Marin Clean Energy (MCE) is a multi-jurisdiction joint-power entity composed of CCAs in several communities in four counties. All communities served by MCE are assigned the same default portfolio of renewable and clean energy use. Thus, if MCE had a 30 percent renewable portfolio that supplied 90 percent of its sales and a 100 percent renewable portfolio that supplied the other 10 percent, each community in MCE would be assigned a renewable energy percentage of 37 percent ($0.9 \times 30 + 0.1 \times 100$).

In some cases, cities within a multi-jurisdiction CCA select different renewable portfolios as their default. When this information is known, each city within a CCA is assigned the renewable and clean energy percentages that correspond to its default portfolio. For example, Calabasas has “Lean Power” as its default option, while Beverly Hills has “100 percent Green Power” as its default option. Then Calabasas is assigned 40 percent renewable, while Beverly Hills is assigned 100 percent. This procedure does not account for the fact that some customers opt for a choice other than the default rate.

C. Logit Regression Results

Based on Chow tests for nonlinearities in the regression equation, I adopt a piecewise linear functional form. The final model indicates that CCA formation is significantly correlated with higher median income, political support for local power, and political support for state regulation of GHG emissions (Table 3). All of these variables have large effects on the probability that a CCA had been formed by 2020. To aid in model interpretation, I assess the relative importance of each significant variable by changing each in isolation within its range of variance in the sample. I construct a hypothetical community with average characteristics for all covariates and determine the model’s prediction of the probability of CCA formation. I then perturb each covariate by one standard deviation, holding all other variables constant, and

measure the difference in predicted probability of CCA formation. An otherwise-average community with income one standard deviation below the sample mean (from \$87,000 to \$44,000) has an estimated probability of CCA formation that is 27.5 percent lower (from 39.7 percent to 12.2 percent). Communities with one standard deviation more support for environmental policy ballot measures are associated with an approximate 25 percentage-point increase in the probability of CCA formation.

Communities with a high modeled probability of forming a CCA tended to be in the actual set of CCA communities as of 2020 (141/165, or 85%, of communities with a propensity score above 50% were CCA communities). As an additional validity check, I look at the overlap between misclassifications and communities that formed CCAs after 2020 (i.e., a time period outside the range used to train the model). There were just thirteen communities for which the model predicted a probability of CCA formation higher than 70%, but which did not actually form CCAs during the observation period. Ten of these thirteen became CCA communities by 2023 (one of the other three was a POU until 2013).

D. Community Characteristics and Decarbonized Energy Procurement

Table 3 demonstrates that communities with higher income and greater support for environmental policy—indicators of higher willingness-to-pay for renewable power—do tend to form CCAs. However, choosing CCA service does not automatically mean higher levels of green energy. Communities may prioritize other potential advantages of the CCA model, such as cost savings or local job creation. Assessing the correlations between community characteristics and decarbonized energy procurement necessitates development of a second regression model. Here, the outcome variable is the percentage of renewable energy in a community’s default portfolio (Table 4). A second version where the outcome variable is the percentage of total carbon-free energy, i.e., renewable plus large hydro and nuclear, can be found in the Appendix. As noted in Section II.A., power providers overseen by local government may be more sensitive to political preferences for green energy relative to private utilities. I therefore compare a simple OLS with a model that has an added public/private LSE fixed effect, so that separate coefficients are obtained for an IOU group versus a pooled CCA and POU group. The CCA and POU group exhibits strong income and scaling effects, meaning that wealthier and larger communities tend to procure more renewable energy. Increasing CCA/POU size by one standard deviation is

associated with a 6.7% increase in renewable energy. Political preferences also appear to be significant determinants of renewable procurement. Stronger Democratic party vote and preference for environmental protection are associated with CCAs and POUs that procure higher levels of renewable energy. A one standard deviation increase in these parameters yields a predicted 4% and 3.7% increase in renewable procurement, respectively.

The pooling of CCAs and POUs reflects their political similarity relative to IOUs. CCAs and POUs are linked to local governments, with each one representing a relatively small set of similar cities. In contrast, the IOU group is dominated by PG&E and SCE, which serve geographically large service territories across dozens of communities. The regressions suggest that for the types of LSEs with local public governance structures—CCAs and POUs—income, political and policy preferences, and scale of the operation are strong predictors of how much that community would likely pursue voluntary green power. These results are in line with intuition: places with a higher willingness to pay for green power are also places leading the charge on voluntary greenness. However, as the CCA/POU pooled group is separate from the IOU group, this may also indicate that the degree of voluntary greenness in CCA/POU communities is quite distinct from that in IOU communities. CCAs are carved out of existing IOU service territory. Although CCAs with high voluntary greenness successfully facilitate matching the groups who most want renewables with the highest levels of renewables, this separation also implies that the residual—the communities remaining with the IOU—have lower preferences and willingness to pay for decarbonization. The net effect on the system will not just depend on the expansion of voluntary greenness. It will also be influenced by the degree to which LSEs representing communities with less interest in decarbonization change their behavior in response to the rise in voluntary greenness. To understand these dynamics, I turn to time series data on actual sales of electricity by fuel type for each LSE.

IV. System Impacts

CCAs are effective in providing more renewable power to their customers; this section places CCAs in the context of overall state performance, where I find that CCA gains have been insufficient to offset deteriorating performance by other types of LSEs. I assemble a panel of administrative data on electric generation sources and sales, and quantify the degree of voluntary overcompliance present for each LSE. I then compare performance trends for CCAs with those

of other LSEs. I also consider performance heterogeneity among CCAs, finding that relatively high-income CCAs procure far more voluntary green power than relatively low-income CCAs do, and large CCAs similarly outperform smaller CCAs. I then expand the analysis from retail sales of power to implications for investment, finding that CCAs largely represent a transfer of renewable energy from existing sources as opposed to new ones.

A. Data and Methodology

Data come from the CEC’s Power Source Disclosure (PSD) program, which is intended to provide consumers “accurate, reliable, and simple to understand information on the sources of energy that are used to provide electric services” (CEC 2024a). These compliance documents are filed by all operational LSEs each year. PSD filings from 2010 through 2020 were obtained via data requests to the CEC ($n = 953$). Each filing lists the quantities procured from generation resources and total retail sales for a given portfolio. Data for 2021–2023 come from a summary of PSD filings published on the CEC website, which contains total retail sales and percentages procured from each generation source for all portfolios. Table 5 lists the number of LSEs of each type that filed PSD documents in recent years. Since 2017, the number of CCAs has tripled.

B. Impacts on System-Wide Performance

This paper is primarily concerned with measuring voluntary overcompliance, so the figures below present values relative to the RPS baseline for a given year as opposed to absolute values. Figure 2 shows performance in terms of procured quantities (TWh) relative to the minimum amounts for compliance with the RPS by LSE type, and Figure 3 shows performance in terms of percentages relative to the RPS percentage in each year. The same data are shown in table form in Appendix Tables A5 and A6. As no new hydroelectric or nuclear projects have been built in recent years, the trends for total clean energy levels are essentially the same as the trends for renewable procurement (see Appendix III for details).

Figures 2 and 3 demonstrate that voluntary over-procurement previously existed in substantial quantities among non-CCA providers, and greater CCA entry was coincident with a shift from over- to under-procurement on the part of other LSEs (Figure 4). CCAs appear to be the only group to consistently exceed mandated levels. After 2018, IOUs no longer consistently procured renewables in excess of the RPS, and the overall state average declined until it reached

the RPS floor. POU performance has generally been near the RPS floor while DAP performance has been mixed. And while fossil fuel use in electricity generation had been declining since 2008, after about 2018, the TWh of electricity generation supplied by fossil fuels plateaued.

Substantial declines in IOU renewable procurement—and disappearance of the state’s voluntary green buffer—are undesirable from a decarbonization policy standpoint. However, these trends are consistent with rational, profit-maximizing behavior. The RPS is a percentage-based standard (renewable MWh divided by total MWh sales). Once load departs an IOU in favor of a CCA, an IOU continuing to hold all its existing contracts becomes over-procured. Because the large IOUs used more renewable energy than the minimum requirement, each could then decrease its quantity of renewable procurement while still being above the state-mandated floor in percentage terms. And since RPS compliance is assessed based on overall performance across a multi-year compliance period (CP), an IOU could even drop under the floor as long as its CP average remained sufficient. Indeed, this appears to be the case for CP3, which spanned 2017 to 2020: IOU performance was well above the RPS target for 2017 and 2018, fell to 2% above the 2019 RPS target, then fell further to 2% below the 2020 RPS target.

Two theoretical explanations for voluntary overcompliance are that it serves as a reputational signal and that it acts as a hedge against future conditions. As shown in Section III, the communities who tend to most value a “green reputation” strongly select into CCA participation. Thus, any incentive for IOUs to signal environmental priorities is attenuated (or possibly even reversed). IOUs may have also over-procured in anticipation of future conditions, as the state has made clear it intends to continue ratcheting up the RPS floor. Consider the case where the IOU expects future supply of renewables to be tight, and therefore RECs will be scarce. Then there is insufficient supply for new CCAs to entirely procure novel resources, so some portion of CCA compliance must come from transferring contracts held by IOUs, presenting an opportunity for the IOU to accrue rents from the resulting transaction. Consider instead the case where the IOU expects future costs to decline, perhaps as a result of CCA voluntary green demand successfully inducing an expansion in renewable supply. Then the IOU would anticipate a lower cost of compliance in the future, making it irrational to continue building a stockpile of renewable procurement. In either case, the IOU’s best short-term response is to jettison its voluntary over-procurement.

The data are consistent with this pattern of CCA growth followed by IOU reduction in voluntary overcompliance. Due to the unusually severe shocks the California electricity system also experienced during this timeframe—the Covid-19 pandemic and the PG&E bankruptcy after the Camp Fire—it is certainly plausible that reductions in IOU performance would have been observed even without CCA entry. Regardless, CCAs’ high level of voluntary green procurement has proven insufficient to stop the erosion of the state’s overall voluntary green buffer.

C. Performance Heterogeneity Among CCAs

IOUs and CCAs clearly differ in their overall levels of renewable procurement, but there is also substantial performance heterogeneity within the set of CCAs. The regression model implies that income and scale strongly affect renewable performance. To examine this as a time series, I separate the set of CCAs into two equally sized groups according to median income (here labeled as “high income” and “low income”). Likewise, CCAs can be divided into two equally sized groups according to MWh of electricity sales in 2020, here labeled as “large” and “small.” I then reproduce Figure 3, except the CCA category is now disaggregated into high and low income, and large and small sales. Both stratifications show a stark difference between the two groups in performance.

Tracing trends in renewable procurement relative to the RPS floor shows that wealthier CCAs succeed in voluntarily procuring renewables in excess of state requirements. However, the degree of overcompliance has declined over time due to both decreases in the percentage of renewables that wealthy CCAs procured and increases in the RPS mandate. In contrast, levels of voluntary greenness have declined substantially for less wealthy CCAs, which now procure levels of renewables close to the RPS floor. By 2021, the less wealthy CCAs procured a lower fraction of renewable electricity than the IOU average (Figure 5). Similar results hold for large versus small CCAs. The renewables share has gradually declined for large CCAs, but the fall has been precipitous for small CCAs. By 2021, small CCAs had a lower renewable share than any other LSE type except for DAPs (Figure 6).

While CCAs overall demonstrate voluntary overcompliance with the RPS mandate, these results emphasize that the performance of individual CCAs is highly variable. Moreover, the gap between larger and smaller (higher- and lower-income) CCAs appears to be growing, with the latter losing ground to the IOUs as suppliers of electricity from renewable sources. As a result,

there are now some CCA customers who receive less green electricity than some IOU customers. The trends suggest that unless CCAs serve wealthier communities or are able to achieve economies of scale—perhaps by aggregating themselves into larger CCAs—then this new model of LSE fails to persistently outperform its IOU counterparts.

V. Customer Inattention and Voluntary Green Participation

Empirical research assessing customer motivations for switching between retailers in deregulated electricity retail settings finds that some customers regard renewable energy content as an important feature and are willing to pay a premium for greener energy. But existing literature also emphasizes the importance of default design (Thaler and Sunstein 2008). In the context of electricity, customers in liberalized retail markets tend to exhibit high levels of inertia, which has made it more difficult for new retailers to attract customers (Hortacsu et al. 2017). Is the CCA model’s success mainly due to deliberate customer action, or due to the opt-out nature of CCAs? As shown in Table 6, by 2021 green “opt-up” offerings were not unique to CCAs. They exist in California for nearly all IOUs and CCAs (as well as for some POUs and one DAP). This translates to approximately 83% of statewide load having access to a voluntary green rate.

When a city joins a CCA, all customers are automatically enrolled in that CCA unless they proactively opt out of CCA service. The city can further select the CCA’s default portfolio of generation sources. Some CCAs only offer portfolios with clean energy content that are significantly above the RPS threshold, meaning 100% of their customers participate in voluntary green power. Further, some cities choose to default all their constituents onto the most expensive and most green option straight away. This unique structure means that the pool of CCA participants actually taking service from a voluntary green rate equals the set of customers who would actively switch to a voluntary green power portfolio, plus all CCA customers on voluntary green service who are too inattentive to switch plans. In most cases, the set of customers opting up from the LSE’s default portfolio to an even cleaner portfolio is quite small. The set of customers who are located in jurisdictions with an automatic 100% clean energy default and who chose to opt down to a less expensive, less green rate, is also small (see Table A11 for detail).

The results on opt-ups and opt-downs reported here are on a per-LSE basis, which does not show how many customers opted to leave CCA service to return to the incumbent IOU. Data obtained from select CCAs indicate that the proportion of opt-outs tends to be less than ten

percent. Opt-out rates are unlikely to be large enough to significantly affect the analysis. Hortacsu et al. (2017) identifies search frictions/inattention and an incumbent provider brand advantage as two major sources of inertia in competitive retail electricity markets. In this setting, CCAs tend to offer either the same rate as the incumbent IOU or moderate cost savings on the order of 5%, and they automatically enroll all customers of member cities, so pricing considerations are unlikely to overcome consumer inattention. Further, any brand loyalty—at least for the CCAs in Northern California—is not likely to lie with beleaguered incumbent utility PG&E, which came in last place in two of the three latest editions of the American Customer Satisfaction Index Energy Utilities Study (ACSI 2025). Overall, then, the large increase in the proportion of customers taking service from voluntary green power portfolios is mainly driven by automatic enrollments rather than active consumer switching.

VI. Implications for Investment

The broader goal of voluntary increases in the demand for renewable electricity is to signal the market to expand the supply of renewable electricity. I explore three measures of CCA impact on the supply of new renewable energy generators: success in procuring novel renewable resources, uptake of long-term contracting, and speed of bringing new construction online.

Even if CCAs procure total renewable energy levels in excess of state mandates, it is not immediately evident whether they also have higher levels of *new* resources in their portfolios than incumbents do. If CCAs seek out new sources of procurement to a larger extent than IOUs do, then they demonstrate an additional dimension to voluntary overcompliance: responsibility for increasing the total renewable capacity of the system. I consider the set of new renewable resources procured in California between 2010 (the first year of CCA operations) and 2020. Each plant in this set can be matched to the retailer(s) that procured power from it in the first year it sold power. This may be different from the allocation of generation in later years—unless 100% of a generator’s capacity is under a long-term contract to a particular LSE, the set of retailers buying its power may change year to year. I calculate that the total amount of added procurement originally attributable to IOUs is 15TWh and the amount originally attributable to CCAs is 5.3TWh. When normalized to annual load, this yields a rate of 0.14 for the IOUs versus 0.11 for CCAs of new resources per unit of power sold. Although CCAs claim a larger share of retail sales from renewable power, much of this turns out to be reallocated from resources originally

procured on behalf of incumbent IOUs such that proportionally, CCAs are responsible for a smaller percentage of new renewable generation sources than IOUs are.

The CPUC's publicly available database of RPS projects includes a list of all long-term contracts signed by IOUs and CCAs through 2021. Long-term contracting is important because it gives greater certainty to renewable developers and is necessary for ensuring sufficient capacity into the future. Figure 7 shows annual cumulative long-term contract quantities signed by IOUs and CCAs respectively. Two key features emerge. First, the rate of total additional capacity under long-term contracts has remained relatively constant, regardless of the rising popularity of CCAs. Second, IOUs virtually stopped signing new long-term contracts in the latter half of the 2010s, instead shifting the responsibility to CCAs. This implies that CCAs do have sufficient capability to contribute to California's long-term contracting needs, but potential net gain has again been offset by IOUs' responses.

The same database can be used to assess how the length of time between initial contract signing and commencement of power delivery has evolved over time. I calculate that CCAs bring new resources online more quickly than incumbents on average, with a mean of 2.5 years between contract execution and commercial online date for long-term contracts signed between 2010 and 2020, as compared to 3.1 years for IOUs. This may be an indication of an important advantage. Even if additionality (in terms of total procurement, or total new procurement) is not evident, the ability to bring resources online sooner would be beneficial for project developers, LSEs, and state planners alike. The large time lags inherent to new generation infrastructure are a challenge for accurate forecasting, and delays create uncertainty around meeting short-term renewable deployment targets. However, Figure 8 shows declining time lags overall, so this may simply be a function of renewable technologies becoming more mature over time. The absence of contemporaneous IOU data in recent years, as noted in Figure 7, precludes direct comparison so it remains unclear whether this is a structural advantage of CCAs or simply a reflection of technological improvements.

VII. Policy Implications and Discussion

California is attempting to achieve total decarbonization of its energy sector by 2045. Overall numbers suggest that the state has been meeting benchmarks and is on schedule. Disaggregating these numbers reveals a more complex picture. Despite the efforts of a select

group of CCAs in procuring very high levels of renewable energy, the state on the whole still hews close to the RPS floor. Communities that have surged ahead in the race to decarbonize should be appropriately recognized for their achievement. However, the fact that these CCA gains occurred alongside undesirable indirect effects for IOU communities—and that such successes have generally not extended to less wealthy or small CCAs—also needs to be recognized.

Power purchases do not exactly reflect the physical realities of the grid. The actual mixture of power delivered at any single instance reflects real-time grid dispatch, which may be very different than what is contracted for. The physics of the power grid dictate that consumers cannot be selective about their electrons. Eventually, if California achieves a grid entirely powered by clean energy resources, then delivered power would indeed be carbon-free at all times and places. Currently, however, a customer who chooses a retail portfolio comprised of 100% renewable energy purchases is still actually receiving electricity generated by the same mixture of generators that serves everyone on California's grid. Thus, the impact of CCAs offering nominally high levels of renewable power procurement is much more modest in a physical grid sense if systemwide levels remain largely unaffected by the behavior of CCAs.

One motivation for forming CCAs was so communities could break away from an IOU, gaining the ability to make their own decisions over power procurement rather than being forced to take the IOU's default power offering. The findings about LSE size imply that scaling up does allow for improved technical capacity. Ironically then, one way for communities to unlock higher levels of renewable energy may be to band back together rather than fracture the retail space further. But a large public entity is not necessarily more responsive to its customers than a large private one: for CCAs, the potential benefits of agglomeration are in tension with the goal of maintaining a local character.

Besides the scaling implications, the regression results suggest that the types of communities most likely to participate in voluntary greenness are well-matched with expectations. These portions of California tend to be wealthier and more politically in favor of environmental causes. Voluntary greenness may therefore be considered a successful mechanism in terms of facilitating the matching of consumers with higher preferences for green energy with higher percentages of green energy actually procured on their behalf. These patterns in voluntary green participation also imply that the remaining group is on average less wealthy with lower

preference for greenness. Moreover, the very success of CCA growth can have important indirect effects on incumbents' behavior. This is clearly illustrated with the departure of CCAs from IOU service territories. IOUs have rationally clustered closer to the RPS floor, attenuating the overall value of CCAs' contributions to the grid. Despite a significant proportion of the state's customers taking service from a voluntary green portfolio, the resulting swell in renewable energy sales is not clearly additional relative to the rising statewide RPS floor.

VIII. Conclusion

The evidence from California sheds light on tactics for enhancing the successful adoption of voluntary greenness as well as potential pitfalls that may weaken its efficacy. For other jurisdictions considering voluntary greenness as a mechanism for decarbonization, the findings impart three key lessons. First, the strategies employed by California's CCAs demonstrate that so-called voluntary greenness might better be thought of as "inattentively green." Active customer switching between default portfolios and voluntary green portfolios is small. Rather than offer opt-in green options, LSEs gained a large portion of customers participating in voluntary greenness by making their default offering a voluntary green portfolio and giving customers the choice to opt back down. Second, the communities most likely to pursue voluntary greenness share certain traits like high income and strong preferences for decarbonization. Voluntary greenness better reflects consumers' preferences by matching portfolios with more decarbonized energy to customers with high willingness to pay for decarbonization. Third, quantifying the benefit of an incremental policy requires correctly identifying the right baseline for comparison. CCAs procure far more than the RPS requires, but excess renewable procurement was also a feature in California before large-scale CCA entry. CCAs' indirect effects on incumbent providers can undermine overall progress. Even in a setting such as California where CCAs serve one-fifth of load, this does not appear to have boosted statewide progress relative to simply enforcing an RPS minimum.

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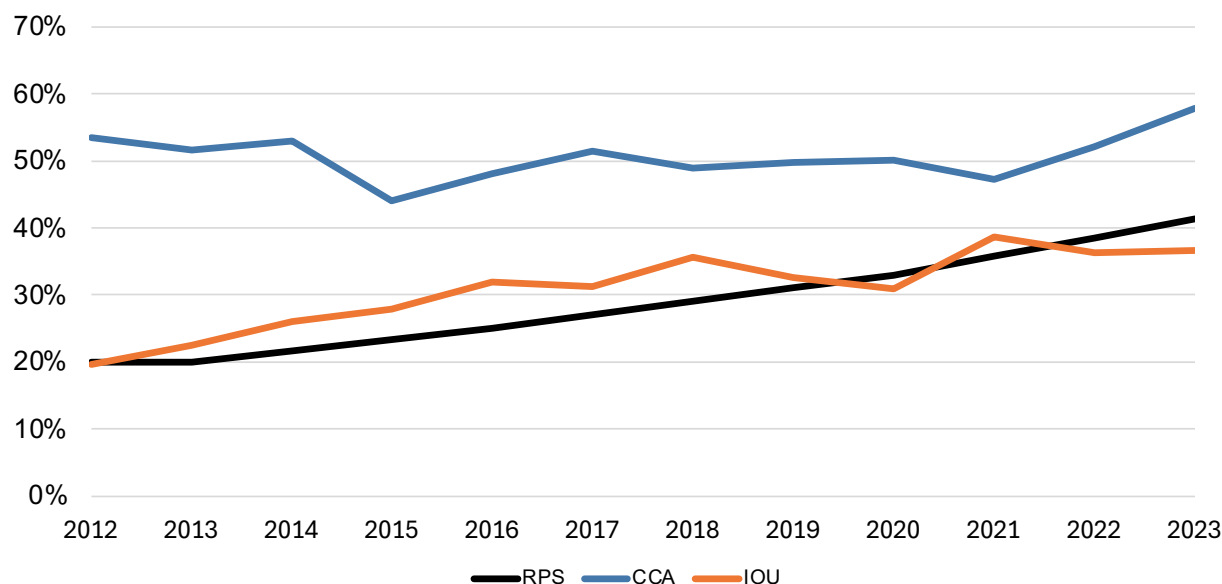
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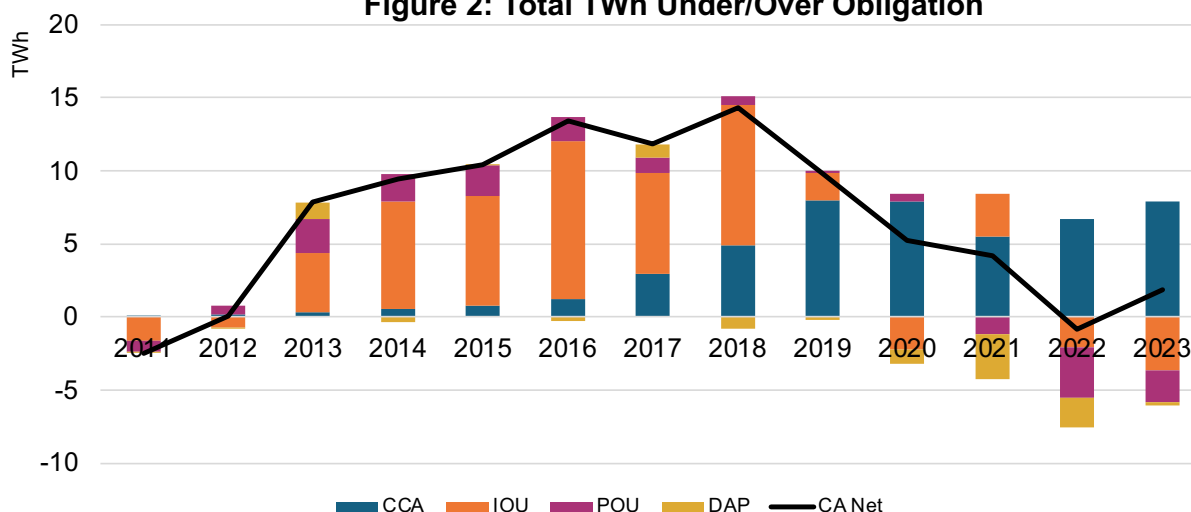
Tables and Figures

Figure 1: Annual Renewable Procurement (%)



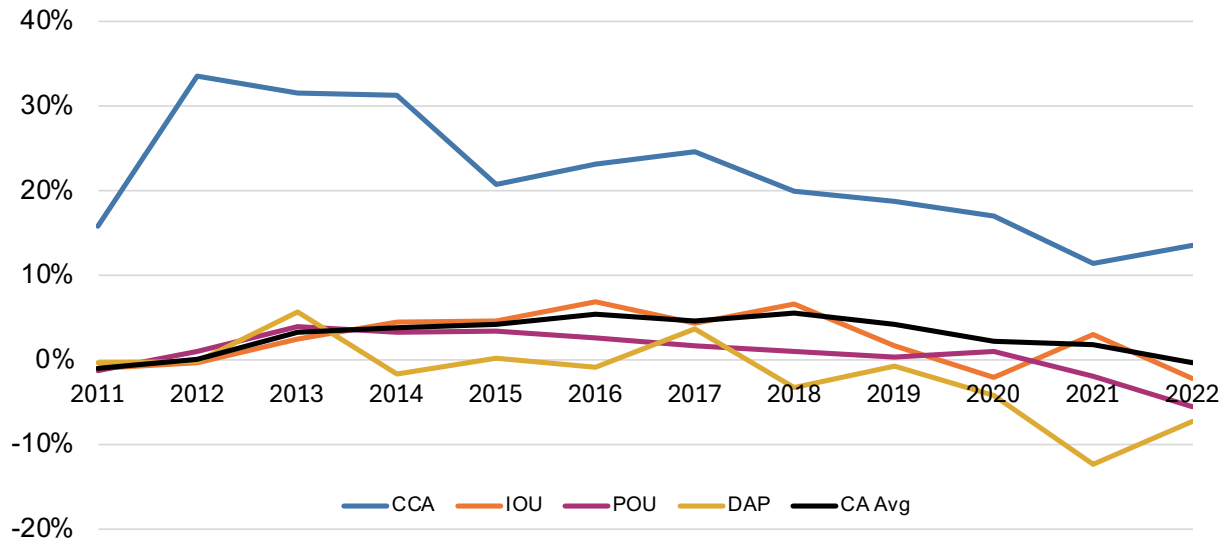
Notes: Graph presents trends in annual average renewable procurement for CCAs and IOUs on a percentage basis, alongside the RPS percentage target for each year. Data are sourced from the California Energy Commission's Power Source Disclosure program.

Figure 2: Total TWh Under/Over Obligation



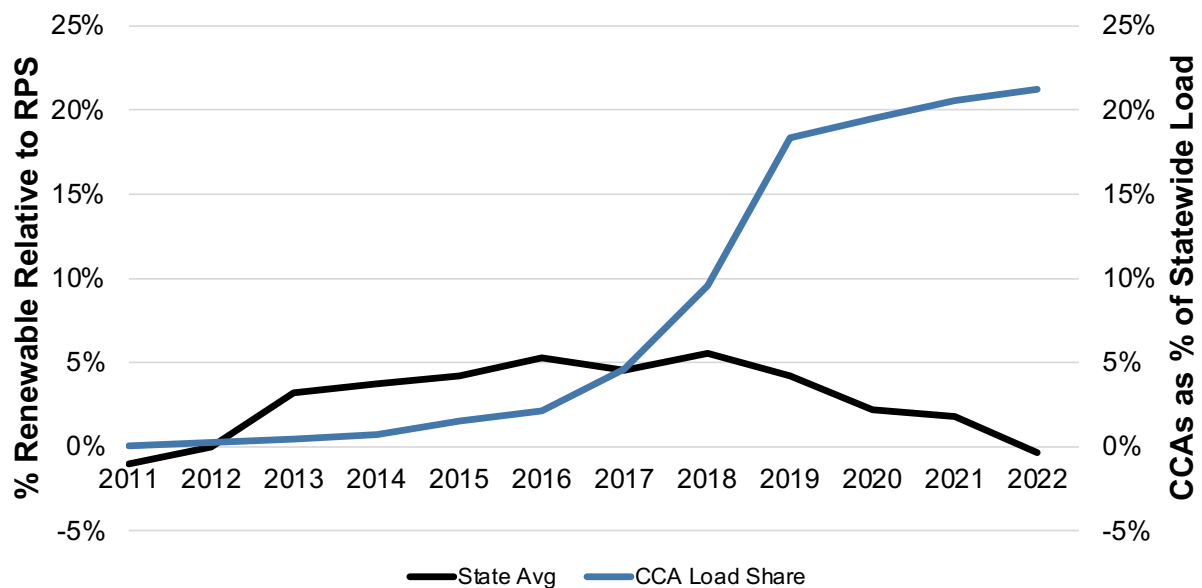
Notes: Graph presents total terawatt-hours under or over the RPS obligation for a given year, by provider type. RPS quantity obligation is calculated as the RPS percentage target times total retail sales in a given year. The black line shows the state's net position. Data are sourced from the California Energy Commission's Power Source Disclosure program.

Figure 3: % Renewable Relative to RPS



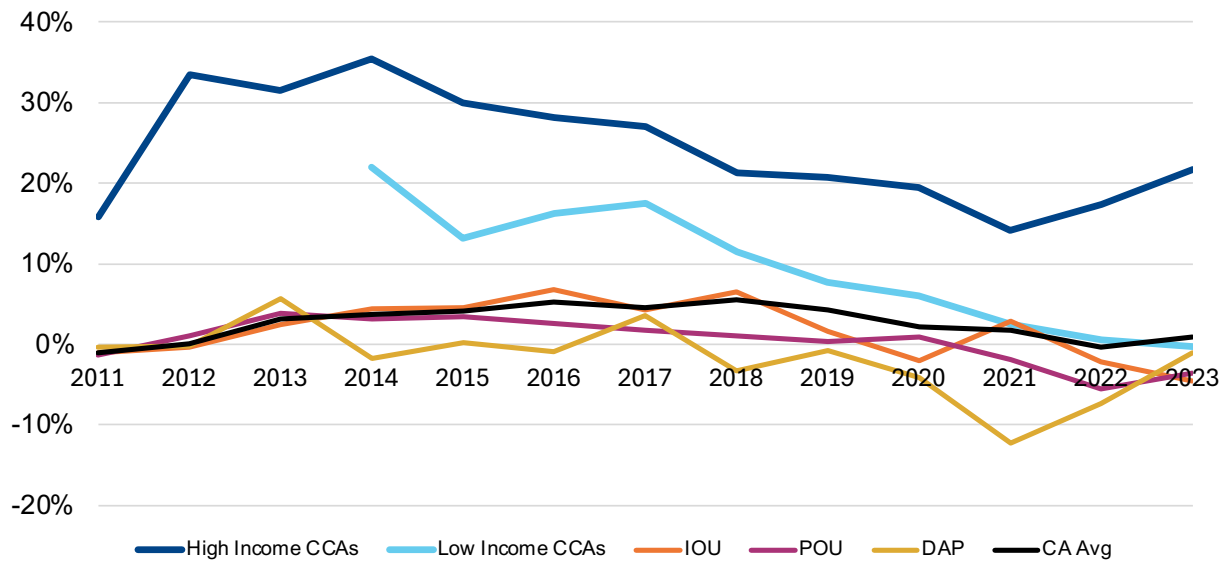
Notes: Graph shows trends in renewable procurement by LSE type relative to the RPS target on a percentage basis. The “CA Avg” line shows the weighted average of all LSEs. Data are sourced from the California Energy Commission’s Power Source Disclosure program.

Figure 4: Trends in CCA Growth and Statewide Overcompliance



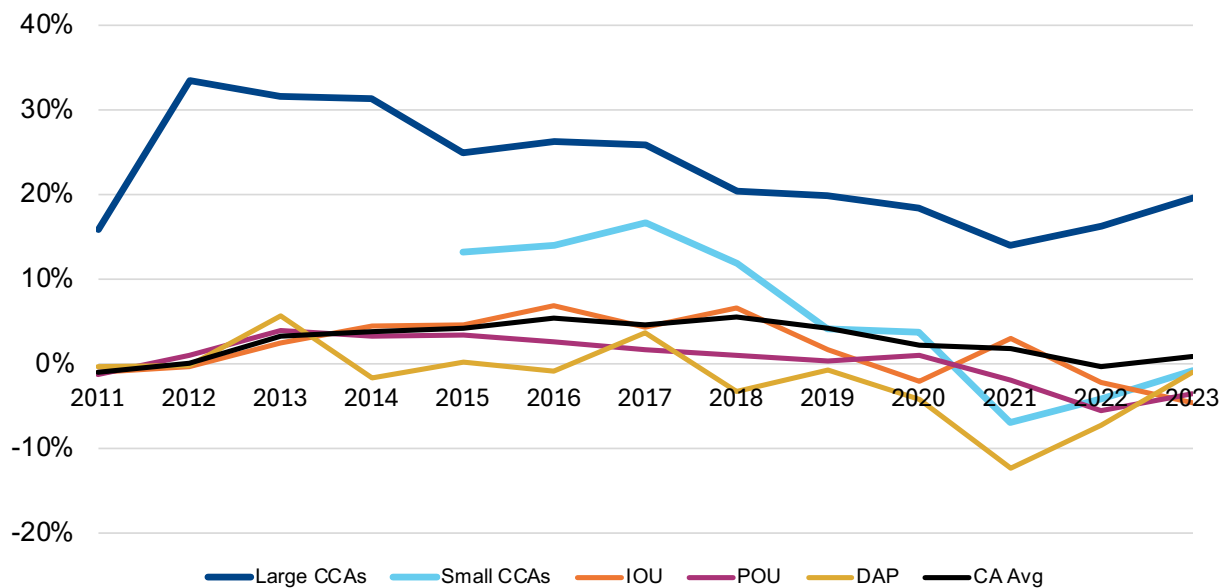
Notes: The black line shows the statewide average level of renewable procurement in excess of the RPS over time. The blue line, plotted on the second axis, shows the proportion of statewide load that is served by CCAs. Data are sourced from the California Energy Commission’s Power Source Disclosure program.

Figure 5: % Renewable Relative to RPS, High- vs Low Income CCAs



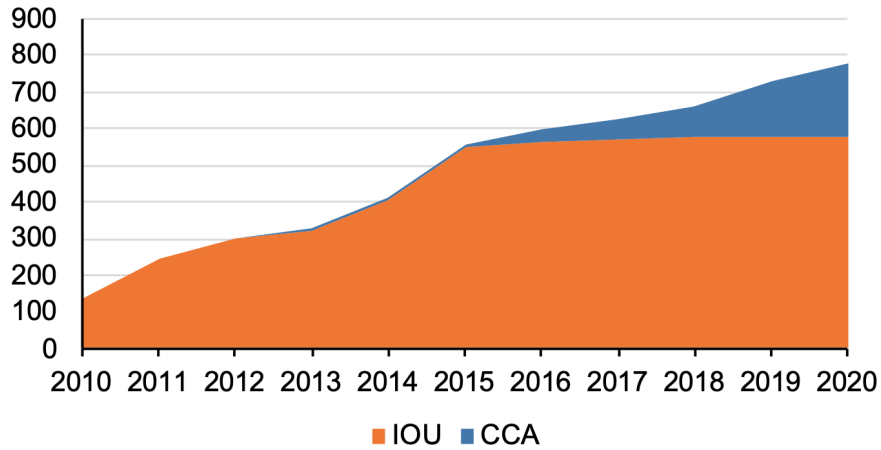
Notes: The chart shows average levels of renewable procurement relative to the RPS percentage. Figure 5 is similar to Figure 3, with the key difference being separation of the CCA set into “High Income” and “Low Income” groups. Data are sourced from the California Energy Commission’s Power Source Disclosure program.

Figure 6: % Renewable Relative to RPS, Large vs Small CCAs



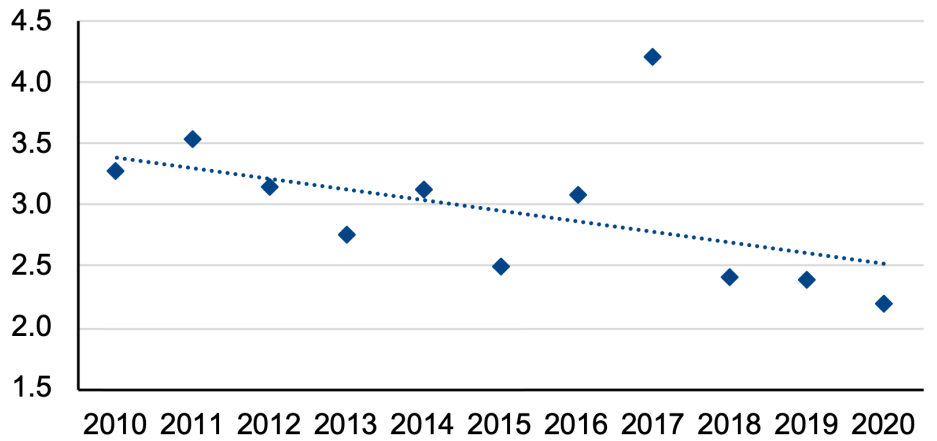
Notes: The chart shows average levels of renewable procurement relative to the RPS percentage. Figure 5 is similar to Figure 3, with the key difference being separation of the CCA set into “Large” and “Small” groups. Data are sourced from the California Energy Commission’s Power Source Disclosure program.

Figure 7: Cumulative TWh of Long-Term Contracts



Notes: Chart shows cumulative terawatt-hours of long-term contracts signed between 2010 and 2020 for IOUs and CCAs. Data come from the California Public Utilities Commission's database of RPS Projects.

Figure 8: Years Between Contract Execution and Commercial Online Date



Notes: Each point represents the average time delay (in years) between contract execution and commercial online dates for publicly-available RPS-eligible projects.

Table 1: Size of each type of LSE in California

	Number of LSEs	Total 2021 Sales (TWh)	Percentage of Total Sales
CCA	25	51	21%
CO-OP	4	0.4	0.2%
DAP	10	24	10%
IOU	6	103	43%
POU	47	60	25%
Total	92	238	100%

Notes: The table provides a descriptive summary of California's five types of Load-Serving Entity: Community Choice Aggregators (CCAs), Member-Owned Cooperatives (Co-Ops), Direct Access Providers (DAPs), Investor-Owned Utilities (IOUs), and Publicly-Owned Utilities (POUs). Data are sourced from the CEC's Power Source Disclosure program.

Table 2: Community Characteristics: Names, Descriptions, Units of Measurement

Variable Name	Description	Units
med_income ¹	Median household income	\$100,000's
pct_white ¹	Percentage identifying as non-Latino White	%
pct_asian ¹	Percentage identifying as Asian	%
pct_some_college ¹	Percentage with at least a high school diploma, some college but no bachelor's	%
pct_bachelors ¹	Percentage with at least a bachelor's	%
pct_democrat ²	Percentage registered with Democratic Party	%
population ¹	Population size	Million people
med_age ¹	Median age	Decades
pct_yes_prop_16 ²	Voting Yes on Prop 16, a ballot measure where yes indicated opposition to local/public power	%
pct_yes_prop_23 ²	Voting Yes on Prop 23, a ballot measure that would have suspended AB 32	%
pct_trump ²	Vote for Donald Trump in the 2020 presidential election	%
pct_manufacturing ¹	Manufacturing share of employment	%
pct_agri ¹	Agricultural share of employment	%
hydro ⁴	Hydro production within same county	MW
pv ⁴	PV production within same city/county	MW
temp_jan ³	Average temperature in January	Tens of deg. F
temp_aug ³	Average temperature in August	Tens of deg. F
lse_size ⁵	LSE size	Billion MWh sales in 2020
not_PGE_SCE	Indicates if the community is outside PG&E or SCE's original service territories	Binary indicator

Notes:

¹ Data for income, race, education, age, population, and employment category come from ACS 5-year tables, 2020 vintage. Census data was pulled with Place (i.e., city/town/CDP) as the granularity, so all CDPs in a given county were aggregated to yield a single value for the unincorporated county.

² Data for political party affiliation, presidential vote, vote on Prop 16, and vote on Prop 23 come from the CA Secretary of State's database. These report a single value for the unincorporated counties so no additional mapping was needed. Six observations were dropped due to missing data for Prop 16 vote.

³ Temperature data come from NOAA Monthly Temperature Normals (1980–2010). Weather stations were matched to cities where this mapping was straightforward (taking simple averages where multiple stations serviced the same city). Otherwise, for smaller cities/towns, the county average value was simply applied.

⁴ Local hydro and local PV production come from the CEC arcGIS data and CEC Energy Almanac. For PV, all CEC-registered solar plants were mapped to the city/unincorporated county where they are physically located. For hydro, all cities within a given county were assigned the value of total hydro production in that county.

⁵ LSE size is based on the total MWh of retail sales for that LSE in 2020, which was calculated by summing up sales across all portfolios offered by each LSE as reported in the 2020 Power Source Disclosure filings. This variable is only used for the regression analyses concerning renewable and carbon-free procurement. It is not included in the logit regression about CCA membership, since the size of a community's LSE is clearly dependent on whether it formed a CCA or not.

Table 3: Propensity of CCA Formation by 2020		
	Linear	Piecewise Linear
med_income	1.035*** (0.394)	3.562*** (0.835)
med_income.seg		-4.734*** (1.229)
pct_yes_prop_16	-6.465*** (1.976)	-9.174*** (2.526)
pct_yes_prop_16.seg		70.890*** (17.775)
pct_yes_prop_23	-9.347*** (1.807)	-9.521*** (2.266)
temp_jan	0.231 (0.316)	0.743* (0.400)
temp_jan.seg		-9.292*** (2.977)
temp_aug	-1.033*** (0.301)	-1.843*** (0.392)
temp_aug.seg		3.721*** (0.913)
not_PGE_SCE	-3.084*** (1.081)	-3.334*** (1.140)
Constant	11.791*** (2.238)	14.547*** (2.884)
Observations	470	470
Log Likelihood	-177.044	-147.417
Akaike Inf. Crit.	368.088	316.835

Notes: I test for structural breaks by identifying one optimal breakpoint, creating indicator variables for the resulting two segments, and re-estimating the generalized linear model to obtain two coefficients per independent variable. Thus, the reported coefficient for the segment above the breakpoint is the difference in slopes for the two segments. For example, the coefficient for median income is 3.562 for communities with median income below the breakpoint of \$108,000 and $3.562 - 4.734 = -1.172$ for communities with median income above \$108,000 (the median household income for the entire sample is \$87,000).

*** p<0.01, ** p<0.05, * p<0.10.

Table 4: Percentage Renewable, Simple OLS vs Public/Private Fixed Effect

	Pooled	[CCAs & POUs] vs [IOUs]
med_income	0.057*** (0.016)	0.074*** (0.018)
med_income.IOU		-0.087*** (0.026)
pct_white	0.049 (0.033)	0.098*** (0.029)
pct_democrat	0.193 (0.144)	0.346*** (0.128)
pct_yes_prop_23	0.119 (0.135)	0.337*** (0.122)
pct_trump	-0.100** (0.050)	-0.205*** (0.074)
pct_trump.IOU		0.179** (0.074)
pct_manufacturing	-0.341** (0.149)	-0.277** (0.132)
temp_jan	0.077*** (0.012)	0.072*** (0.019)
temp_jan.IOU		-0.029 (0.023)
lse_size	-3.269*** (0.313)	19.617*** (2.246)
lse_size.IOU		-20.221*** (2.229)
not_PGE_SCE	-0.168*** (0.017)	-0.095*** (0.017)
is_IOU		0.116 (0.116)
Constant	-0.031 (0.133)	-0.279* (0.145)
Observations	523	523
Log Likelihood	379.514	452.059
Akaike Inf. Crit.	-739.028	-874.118

Notes: The first column is an OLS of pct_renewable against various community-level characteristics. The second column includes a public-private fixed effect for communities taking service from public providers (CCAs and POUs) versus investor-owned utilities. *** p<0.01, ** p<0.05, * p<0.10.

Table 5: Number of LSEs with PSD filings by year

	2017	2018	2019	2020	2021	2022	2023
CCA	9	19	19	22	25	25	25
CO-OP	2	4	4	4	4	4	4
DA	14	14	13	13	10	10	10
IOU	6	6	6	6	6	6	6
POU	44	46	46	47	47	47	47
Total	75	89	88	92	92	92	92

Notes: Table shows the number of active load-serving entities reporting to the California Energy Commission each year. Values prior to 2017 are not displayed because few CCAs were operational before then, and no other meaningful entries or exits occurred. Since 2017, the number of CCAs has nearly tripled and the total number of providers in California has increased by over 20%. Data are sourced from the CEC's Power Source Disclosure program.

Table 6: California LSEs Offering Choice of Multiple Procurement Portfolios, 2021

	Number of LSEs	Number Offering Portfolio Choice	Percentage (by load) Offering Portfolio Choice
IOU	6	4	99.3%
CCA	25	24	99.9%
POU	47	13	75.2%
DA	10	1	2.2%
CO-OP	4	0	0.0%

Notes: Table shows the number of load-serving entities offering portfolio choice as of 2021. Portfolio choice is defined as the retailer offering at least two plans that contain differing levels of renewable energy. Because LSEs are highly heterogeneous in size, the right-hand column displays the percentage by load share offering portfolio choice for each category. Taking the weighted average of all provider types yields 83% of California customers having access to portfolio choice. Data are sourced from the CEC's Power Source Disclosure program.

Appendix

Appendix I. Chow Tests for Structural Breaks

Because the functional form of the underlying equation cannot be derived from theory, a different functional form plausibly may be a better representation of the underlying relationship. To test whether the functional form is non-linear, piecewise linear specifications were estimated in which one of the continuous covariates is divided into two segments while leaving all other variables unchanged. Segmenting the data adds an additional variable, a tradeoff that is worthwhile if the resulting piecewise specification adds explanatory power. This was determined by using ANOVA (Table A1). The variables found to improve explanatory power when segmented were added on to the original binomial logit. Then, variable selection was performed using a standard backwards stepwise procedure to eliminate the worst-performing terms. This yielded the specification shown in Table 3 of the main text.

Table A1: Alternative Piecewise Specifications by Explanatory Variable

Parameter	Pr(> χ)
med_income	9.79E-04 ***
pct_white	0.002 **
pct_asian	0.086 .
pct_some_college	0.079 .
pct_bachelors	0.086 .
pct_democrat	0.164
population	0.058 .
med_age	0.454
pct_yes_prop_16	2.61E-04 ***
pct_yes_prop_23	0.162
pct_trump	0.019 *
pct_manufacturing	0.220
pct_agri	0.156
hydro	0.130
pv	0.670
temp_jan	9.13E-05 ***
temp_aug	6.74E-04 ***

Notes: Table entries summarize ANOVA results of testing structural breaks for each variable at a time. Statistically significant values indicate that segmenting the variable adds explanatory power, and so the variable should be treated as piecewise linear in the full specification. *** p<0.01, ** p<0.05, * p<0.10.

Appendix II. Pooling of Public Versus Privately Governed Retailers

I test the validity of the private/public fixed effect by comparing an unpooled specification to one in which the CCAs and POU are pooled. Differences in the significance and magnitude of coefficients between the IOU, CCA, and POU groups are detected by interacting each covariate with each LSE type, and conducting a Wald test to assess whether the equations pool (Tables A2 and A3). The POU and CCA equations for renewable energy appear to pool, but are distinct from the IOU group. All three groups appear distinct across some of the variables for the total clean energy regressions. This is likely because several POUs have large quantities of hydroelectric assets (i.e., high levels of clean power), which is a function of their historical development as opposed to the proxies for willingness-to-pay for clean energy that I test for.

Table A2: Probability that the difference between the coefficients for the two groups is indistinguishable from zero (for the pct_renewable regressions)

parameter name (i)	$\Pr(>\chi^2)$ for test " $\beta_{i,CCA} - \beta_{i,IOU} = 0$ "	$\Pr(>\chi^2)$ for test " $\beta_{i,POU} - \beta_{i,IOU} = 0$ "	$\Pr(>\chi^2)$ for test " $\beta_{i,CCA} - \beta_{i,POU} = 0$ "
med_income	6.66E-05 ***	0.023 *	0.646
pct_white	0.542	0.292	0.517
pct_asian	0.241	0.255	0.070 .
pct_some_college	1.19E-04 ***	0.062 .	0.663
pct_bachelors	9.62E-05 ***	0.016 *	0.963
pct_democrat	0.021 *	0.977	0.123
population	0.868	0.571	0.651
med_age	0.025 *	0.986	0.292
pct_yes_prop_16	0.815	0.246	0.293
pct_yes_prop_23	0.015 *	0.929	0.066 .
pct_trump	7.63E-04 ***	0.311	0.151
pct_manufacturing	0.096 .	0.911	0.151
pct_agri	0.587	0.361	0.629
hydro	8.39E-03 **	0.600	0.272
pv	0.072 .	0.588	0.070 .
temp_jan	1.30E-05 ***	1.44E-03 **	0.346
temp_aug	0.209	0.937	0.299
lse_size	2.59E-21 ***	0.018 *	5.48E-03 **

Notes: Table entries represent the probability that coefficients between two of the groups are equal, which would indicate that they come from the same underlying distribution. Statistically significant values indicate that the data cannot be pooled, i.e., the form of the LSE matters in governing the relationship between community characteristics and proportion of renewables procured.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A3: Probability that the difference between the coefficients for the two groups is indistinguishable from zero (for the pct_clean_energy regressions)

parameter name (i)	$\Pr(>\chi^2)$ for test " $\beta_{i,CCA} - \beta_{i,IOU} = 0$ "	$\Pr(>\chi^2)$ for test " $\beta_{i,POU} - \beta_{i,IOU} = 0$ "	$\Pr(>\chi^2)$ for test " $\beta_{i,CCA} - \beta_{i,POU} = 0$ "
med_income	9.19E-05 ***	0.524	0.243
pct_white	0.718	5.19E-09 ***	1.10E-07 ***
pct_asian	0.217	9.34E-03 **	8.86E-04 ***
pct_some_college	6.87E-05 ***	0.260	4.18E-04 ***
pct_bachelors	1.02E-04 ***	0.090 .	0.517
pct_democrat	6.47E-04 ***	6.52E-04 ***	4.13E-08 ***
population	0.297	0.743	0.322
med_age	0.037 *	5.68E-03 **	0.101
pct_yes_prop_16	0.112	0.260	0.963
pct_yes_prop_23	6.08E-05 ***	0.197	3.37E-05 ***
pct_trump	1.54E-05 ***	0.010 *	6.10E-08 ***
pct_manufacturing	0.890	0.068 .	0.063 .
pct_agri	0.923	0.515	0.605
hydro	2.18E-03 **	0.461	0.247
pvc	0.022 *	0.873	0.060 .
temp_jan	0.010 **	0.312	4.37E-03 **
temp_aug	6.70E-04 ***	0.035 *	0.276
lse_size	7.02E-03 **	8.14E-07 ***	4.62E-03 **

Notes: Table entries represent the probability that coefficients between two of the groups are equal, which would indicate that they come from the same underlying distribution. Statistically significant values indicate that the data cannot be pooled, i.e., the form of the LSE matters in governing the relationship between community characteristics and proportion of clean energy procured.

*** p<0.01, ** p<0.05, * p<0.10.

I also look at the association between the total percentage of clean energy as a function of community characteristics (Table A4); this is analogous to Table 4 in the main text except the outcome variable is total clean energy (i.e., includes nuclear and hydro) rather than renewable energy.

Table A4: Percentage Total Clean Energy Coefficients, Simple OLS vs Unpooled

	Pooled	Unpooled
med_income	0.059** (0.024)	
med_income.CCA		0.106*** (0.021)
pct_white	-0.002 (0.040)	
pct_white.POU		0.889*** (0.136)
pct_asian	-0.146** (0.068)	-0.103** (0.045)
pct_yes_prop_23	-0.059 (0.104)	
pct_yes_prop_23.POU		1.185*** (0.415)
pct_trump	-0.125* (0.068)	-0.058 (0.049)
pct_trump.CCA		-0.495*** (0.100)
pct_trump.POU		-0.928*** (0.340)
hydro	-0.026*** (0.010)	
hydro.CCA		-0.133*** (0.019)
temp_jan	-0.066*** (0.017)	0.055*** (0.014)
lse_size	-5.740*** (0.456)	-17.083*** (0.815)
lse_size.CCA		20.569*** (4.038)
lse_size.POU		32.929*** (5.261)
not_PGE_SCE	-0.407*** (0.025)	-0.958*** (0.038)
is_POU		-0.487*** (0.116)
is_CCA		-0.555*** (0.057)
Constant	1.255*** (0.092)	1.195*** (0.057)
Observations	523	523
Log Likelihood	189.577	348.086
Akaike Inf. Crit.	-359.154	-664.172

Appendix III. Time Trends in the Percentages of Renewable and Clean Energy Procured

Table A5: TWh Annual Procurement by LSE Type, Relative to Minimum RPS Quantity

	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
CCA	0.0	0.2	0.4	0.6	0.8	1.2	3.0	4.9	8.0	7.9	5.5	6.7
IOU	-1.6	-0.7	4.0	7.3	7.5	10.8	6.9	9.6	1.8	-2.2	2.9	-2.1
POU	-0.8	0.6	2.3	1.9	2.1	1.6	1.1	0.6	0.2	0.6	-1.1	-3.4
DAP	-0.1	0.0	1.1	-0.4	0.0	-0.2	0.9	-0.8	-0.2	-1.0	-3.1	-2.0
CA Net	-2.5	0.1	7.8	9.4	10.4	13.4	11.8	14.3	9.8	5.2	4.2	-0.8

Table A6: Annual renewable procurement by LSE type relative to RPS target (%)

	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
CCA	16%	33%	32%	31%	21%	23%	24%	20%	19%	17%	11%	14%
IOU	-1%	0%	2%	4%	5%	7%	4%	7%	2%	-2%	3%	-2%
POU	-1%	1%	4%	3%	3%	3%	2%	1%	0%	1%	-2%	-6%
DAP	0%	0%	6%	-2%	0%	-1%	4%	-3%	-1%	-4%	-12%	-7%
CA Avg	-1%	0%	3%	4%	4%	5%	5%	6%	4%	2%	2%	0%

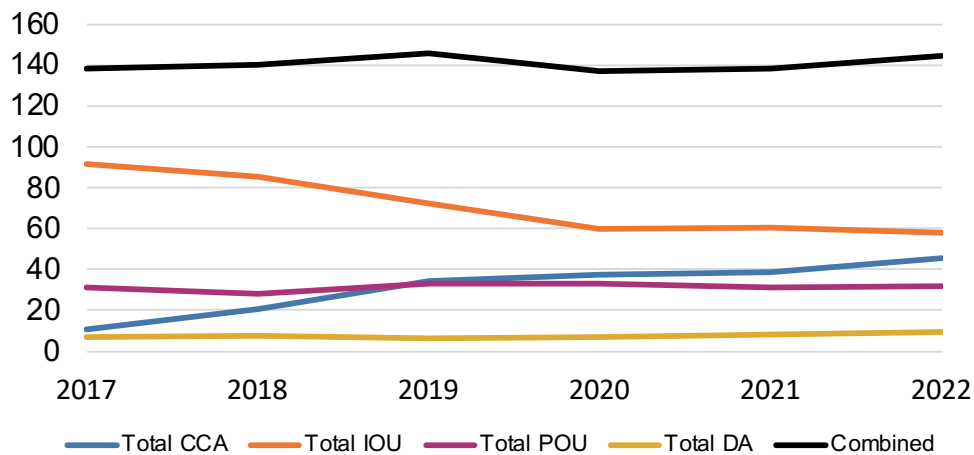
Table A7: Trends in Renewable Procurement Relative to the RPS for High versus Low Income, and Large versus Small, CCAs

	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
High Income CCA	16%	33%	32%	35%	30%	28%	27%	21%	21%	20%	14%	17%
Low Income CCA	N/A	N/A	N/A	22%	13%	16%	18%	11%	8%	6%	2%	1%
Large CCA	16%	33%	32%	31%	25%	26%	26%	20%	20%	18%	14%	16%
Small CCA	N/A	N/A	N/A	N/A	13%	14%	17%	12%	4%	4%	-7%	-4%
IOU	-1%	0%	2%	4%	5%	7%	4%	7%	2%	-2%	3%	-2%
POU	-1%	1%	4%	3%	3%	3%	2%	1%	0%	1%	-2%	-6%
DAP	0%	0%	6%	-2%	0%	-1%	4%	-3%	-1%	-4%	-12%	-7%
CA Avg	-1%	0%	3%	4%	4%	5%	5%	6%	4%	2%	2%	0%

Table A8: Total TWh of Clean Energy procurement in a given year, by provider type

	2017	2018	2019	2020	2021	2022
Total CCA	10.5	20.5	34.5	37.3	38.6	45.2
Total IOU	91.8	85.2	72.2	59.9	60.2	58.0
Total POU	30.9	27.9	32.9	33.3	31.4	31.8
Total DA	6.6	7.2	6.0	6.8	8.2	9.7
Combined	138.5	139.9	145.6	137.2	138.4	144.8

Figure A1: Clean Energy Procurement (TWh)

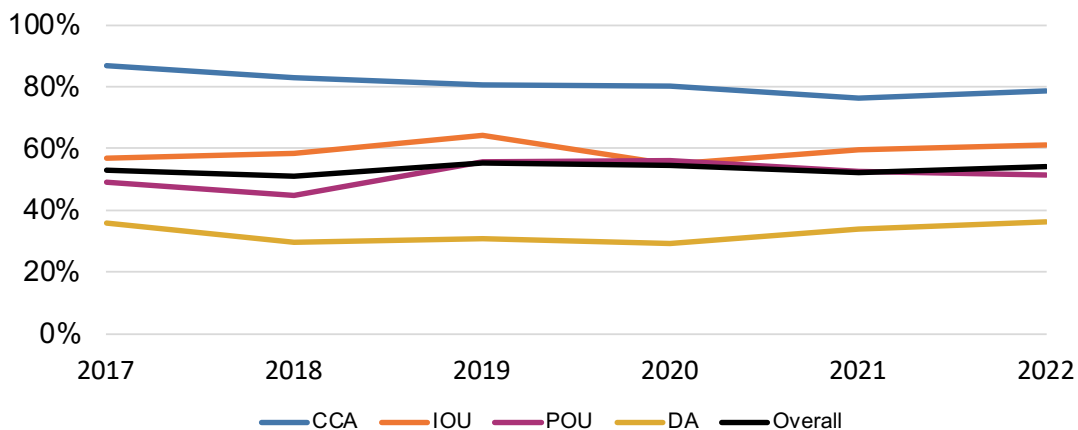


Note: Figure shows the total TWh of clean energy procurement in a given year, by provider type. “Clean energy” is the sum of renewable, large hydro, and nuclear energy. Since there is no particular mandated percentage of clean energy, values are reported as absolute TWh, instead of quantities relative to a compliance floor.

Table A9: Annual total clean energy procurement by LSE type (%)

	2017	2018	2019	2020	2021	2022
CCA	87%	83%	81%	80%	76%	79%
IOU	57%	58%	64%	55%	59%	61%
POU	49%	45%	56%	56%	53%	51%
DA	36%	30%	31%	29%	34%	36%
Overall	53%	51%	55%	55%	52%	54%

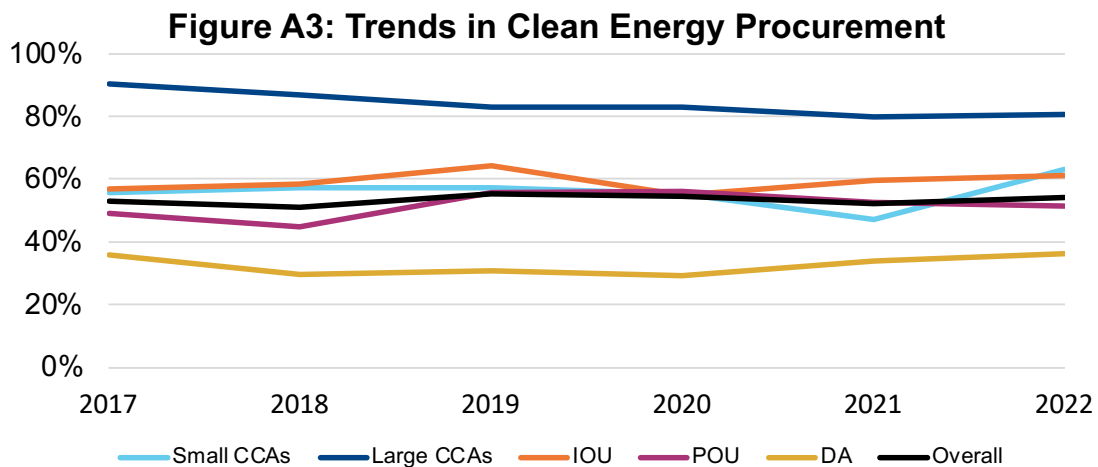
Figure A2: Trends in Clean Energy Procurement



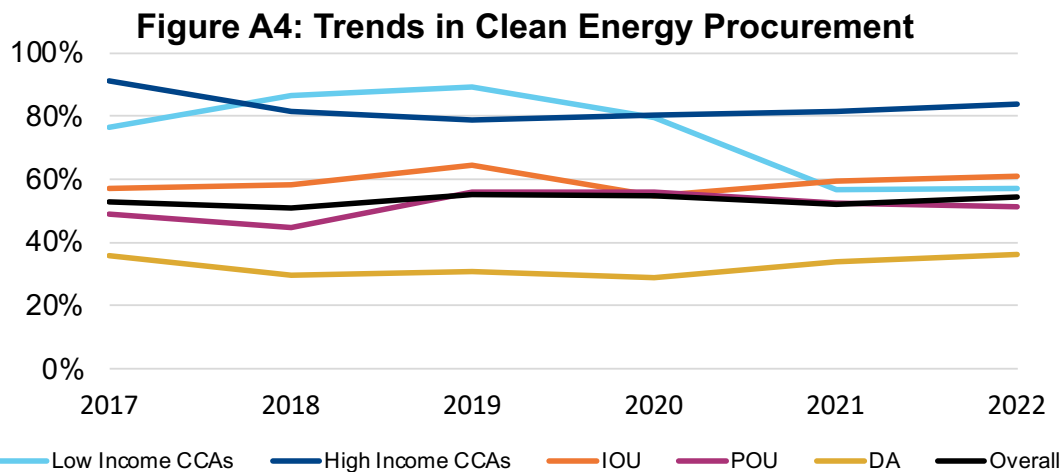
Note: Figure shows trends in total clean energy procurement by LSE type, expressed as a percentage of overall portfolio. “Clean energy” is the sum of renewable, large hydro, and nuclear energy. Since there is no particular mandated percentage of clean energy, values are reported as actual percentages, instead of percentages relative to a compliance floor.

Table A10: Trends in Total Clean Energy Procurement Relative to the RPS for High versus Low Income, and Large versus Small, CCAs

	2017	2018	2019	2020	2021	2022
High Income CCA	91%	81%	79%	80%	81%	84%
Low Income CCA	76%	87%	89%	80%	57%	57%
Large CCA	90%	87%	83%	83%	80%	81%
Small CCA	56%	57%	57%	55%	47%	63%
IOU	57%	58%	64%	55%	59%	61%
POU	49%	45%	56%	56%	53%	51%
DA	36%	30%	31%	29%	34%	36%
Overall	53%	51%	55%	55%	52%	54%



Note: Figure A3 reproduces Figure A2, with the key difference being separation of the CCA set into “Large” and “Small” groups.



Note: Figure A4 reproduces Figure A2, with the key difference being separation of the CCA set into “High Income” and “Low Income” groups.

Appendix IV. Detail on Opt-Up Rates by LSE

Table A11: Opt-Up Rates for California LSEs Offering Multiple Portfolios, 2021

LSE	Type	% on Least Green Plan	% on Greener Plans
Apple Valley Choice Energy	CCA	99.8%	0.2%
Baldwin	CCA	100.0%	0.0%
Central Coast Community Energy (3CE)	CCA	99.2%	0.8%
Clean Energy Alliance	CCA	1.3%	98.7% ¹
Clean Power Alliance of SoCal	CCA	21.6%	78.4% ²
CleanPowerSF	CCA	93.9%	6.1%
Desert Community Energy	CCA	9.7%	90.3% ¹
East Bay Community Energy	CCA	83.3%	16.7% ²
Lancaster Choice Energy	CCA	99.1%	0.9%
Marin Clean Energy	CCA	96.4%	3.6%
Peninsula Clean Energy	CCA	91.8%	8.2% ³
Pico Rivera Innovative Municipal Energy	CCA	97.3%	2.7%
Pioneer Community Energy	CCA	100.0%	0.0%
Pomona	CCA	100.0%	0.0%
Rancho Mirage Energy Authority	CCA	99.4%	0.6%
Redwood Coast Energy Authority	CCA	99.1%	0.9%
San Diego Community Power	CCA	93.7%	6.3%
San Jacinto Power	CCA	99.9%	0.1%
San Jose Clean Energy	CCA	96.6%	3.4%
Santa Barbara Clean Energy	CCA	12.7%	87.3% ¹
Silicon Valley Clean Energy	CCA	96.6%	3.4%
Solana Energy Alliance	CCA	99.1%	0.9%
Sonoma Clean Power Authority	CCA	96.6%	3.4%
Valley Clean Energy Alliance	CCA	99.4%	0.6%
3 Phases Renewables	DA	1.0%	99.0%
PG&E	IOU	98.0%	2.0%
PacifiCorp	IOU	98.8%	1.2%
SDG&E	IOU	99.6%	0.4%
SCE	IOU	99.9%	0.1%
Alameda Municipal Power	POU	95.2%	4.8%
Burbank Water and Power	POU	99.9%	0.1%
City of Healdsburg Electric Utility	POU	91.5%	8.5% ⁴
CCSF	POU	93.4%	6.6%

LADWP	POU	99.8%	0.2%
City of Palo Alto Utilities	POU	97.0%	3.0%
City of Pasadena	POU	95.2%	4.8%
PWRPA	POU	88.4%	11.6%
City of Riverside Public Utilities	POU	99.9%	0.1%
City of Roseville	POU	99.8%	0.2%
Silicon Valley Power	POU	97.0%	3.0%
SMUD	POU	90.9%	9.1%
Turlock Irrigation District	POU	100.0%	0.0%

Notes:

1. All communities in this CCA were automatically enrolled in the most premium, greenest rate. Thus, “% on Least Green Plan” represents active opt-downs from the default option.
2. Some communities in this CCA were automatically enrolled in the most premium, greenest rate. However, data on the breakdown of load by member community are unavailable.
3. One community in this CCA was automatically enrolled in the most premium, greenest rate.
4. Municipal electricity use was automatically enrolled in the most premium, greenest rate.

Source: CEC Power Source Disclosure program